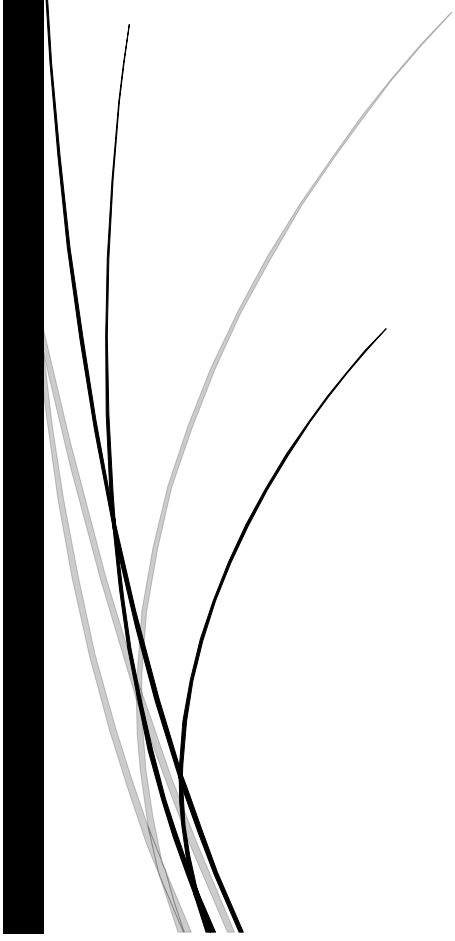
Bayesian Networks for Heart Disease Prediction

Assignment 3

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Introduction

The objective was to build and compare two Bayesian networks for heart-disease prediction on the Cleveland dataset: an expert-designed DAG with twenty-one edges, and a DAG of eleven edges learned by hill-climbing under the Bayesian Information Criterion. Both were trained on the same 242 patients and tested on the same 61 held-out patients, so any difference between them reflects structure rather than data.

On the held-out test set, the two networks perform comparably rather than one clearly outperforming the other. The expert network ranks patients slightly better (ROC-AUC 0.939 versus 0.918); the learned network is more accurate at the 0.50 threshold (86.9% versus 85.2%), with fewer false positives. A paired bootstrap of the AUC difference gives a 95% interval of −0.098 to 0.039, which spans zero, so neither network is proven superior. The Thalassemia result, chest-pain type, and the number of vessels affected are the strongest predictors in both networks and alone form the learned network's Markov blanket.

The number of vessels affected is the single most influential variable once other evidence is fixed, producing the largest shift in predicted risk of any variable tested. The learned network suits cases needing a compact, well-supported model; the expert network suits cases needing traceable clinical reasoning and better ranking. Two learned edges run from a test finding into age or sex, which is not biologically plausible and should be read as a search artefact rather than a causal claim. Any practical use would require validation on independent patients and a review of the model's reliance on the invasive vessel-count test.

1. Data Collection and Preparation

The dataset contains 303 patients and fourteen columns. Schema, category codes, and duplicates were verified by assertion; no duplicate rows were found. Six cells were missing: four in the vessels-affected variable and two in the thalassemia result and were treated as missing rather than zero. The column gender was renamed sex to match the assignment's coding.

The five-level outcome was converted to a binary target, heartdisease_bin, and removed from the predictor set to avoid target leakage. Of 303 patients, 139 (45.9%) have disease, a mild imbalance that did not warrant resampling; precision and recall

are reported alongside accuracy. A stratified 80/20 split with a fixed seed gave 242 training and 61 test patients at matching prevalence. All imputation values and discretisation cut points were computed on the training set only and applied unchanged to the test set.

Table 1. Patient counts and disease prevalence in the full dataset, the training set and the test set.

Split	n	No disease	Disease	Disease %
All patients	303	164	139	45.9
Training set	242	131	111	45.9
Test set	61	33	28	45.9

1.1 Discretisation

A discrete Bayesian network requires each variable to take a small number of states, so the five continuous measurements were converted into categorical bands. Age, resting blood pressure, cholesterol and maximum heart rate were each divided into three roughly equal-sized groups, labelled Low, Medium and High, using the tertiles of their training-set distributions. The ST-depression variable, ‘oldpeak’, was treated differently because many patients recorded no depression at all: a value of zero was kept as its own category, and the remaining positive values were split at their training-set median into Moderate and High bands. These bands are data-relative, not clinical thresholds.

Table 2. Discretisation cut points and resulting categories for the five continuous variables.

Variable	Cut points	Resulting categories
Age	51 years, 59 years	Low / Medium / High
Resting blood pressure	120, 139.3 mm Hg	Low / Medium / High
Cholesterol	226, 266.7 mg/dl	Low / Medium / High

Variable	Cut points	Resulting categories
Maximum heart rate	143, 162 beats/min	Low / Medium / High
ST depression (oldpeak)	0, 1.2	No depression / Moderate / High

1.2 Explanatory analysis and feature selection

All thirteen predictors were retained so both structures compete on identical information. The predictors were ranked by their univariate mutual information with the target places the thalassemia result first (0.156 nats), followed by chest-pain type (0.127), the number of affected vessels (0.116), exercise-induced angina (0.091), maximum heart rate (0.073) and the ST-segment slope (0.071); fasting blood sugar (0.001), resting blood pressure (0.008) and cholesterol (0.014) carry almost no univariate association with the outcome on their own. These figures were read as a preliminary indication of which variables are likely to matter, and not as a formal test of significance because mutual information captures association rather than causal or independent predictive contribution.

2. Structured Learning

2.1 The expert- designated network

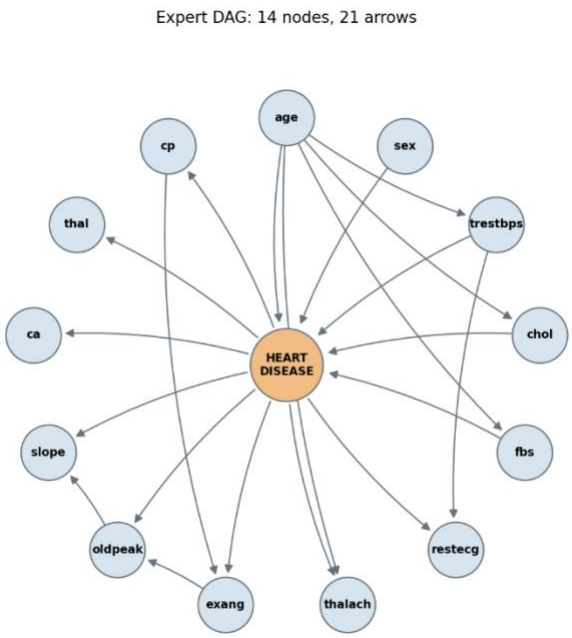


Figure 1. The expert-designated directed acyclic graph, comprising fourteen nodes and twenty-one directed edges, with the disease node placed at the centre. The expert network was constructed so that a patient's age and their baseline cardiovascular risk factors, namely blood pressure, cholesterol, fasting blood sugar and sex, each influence the probability of disease. While disease in turn influences the

symptoms and diagnostic test findings that a clinician would observe. Observing any of these downstream findings still updates the estimated probability of disease, even though the corresponding edge points go from disease toward the finding rather than the reverse. This is because the exact inference in a Bayesian network is not restricted to the direction in which its edges are drawn. The connections represent modelling judgements grounded in clinical reasoning; they are not claims, proven by the dataset, that the underlying relationships are causal.

Three pathways feed the target: age acts directly; each of the four baseline risk factors (blood pressure, cholesterol, fasting glucose, sex) acts directly; and age also acts indirectly, through those risk factors. With five parents, the target's table spans 108 configurations

Table 3. Justification provided for each of the 21 edges in the expert- designated network.

Parent variable	Child variable	Justification
Age	Resting blood pressure	Blood pressure tends to rise with age.
Age	Fasting blood sugar	Glucose regulation tends to decline with age, although this indicator is not itself a diabetes diagnosis.
Age	Cholesterol	Cholesterol levels vary across age groups, without a necessarily monotonic pattern.
Age	Disease	Age provides baseline information about the probability of disease.
Age	Maximum heart rate	Maximum achievable heart rate declines with age, a relationship documented by Tanaka, Monahan and Seals (2001).
Resting blood pressure	Disease	Hypertension is an established cardiovascular risk factor.
Resting blood pressure	Resting ECG	Elevated blood pressure can produce ventricular hypertrophy, which is detectable on a resting ECG.
Fasting blood sugar	Disease	An elevated fasting glucose reading serves as a limited proxy for metabolic risk.
Cholesterol	Disease	Cholesterol is a recognised baseline cardiovascular risk factor.
Sex	Disease	Baseline disease prevalence differs between male and female patients.

Parent variable	Child variable	Justification
Disease	Maximum heart rate	The presence of disease can limit the maximum heart rate a patient is able to achieve during exercise.
Disease	Chest-pain type	The type of chest pain reported is itself a diagnostic finding whose distribution depends on disease status.
Disease	Resting ECG	Resting ECG abnormalities can differ by disease status, although a normal ECG does not itself rule out disease.
Disease	Thalassemia result	Perfusion-test findings carry diagnostic information about the presence of disease.
Disease	Vessels affected	Fluoroscopic vessel counts are linked to the diagnostic investigation of disease, though this test is not used for general screening.
Disease	Exercise angina	Ischaemia associated with disease can provoke angina during exercise.
Disease	ST depression	Ischaemia associated with disease can produce measurable ST-segment depression during exercise.
Disease	ST slope	Ischaemia associated with disease can alter the slope of the ST segment during exercise.
Chest-pain type	Exercise angina	Chest-pain type and exercise-induced angina are related symptoms, and the direction chosen here reflects a modelling choice about how to factorise their relationship.
Exercise angina	ST depression	Angina and ST depression are both observations made during the same exercise test, and their relationship is treated as an assumption rather than an established causal link.
ST depression	ST slope	The magnitude of ST depression and the slope of the ST segment are both derived from the same exercise response, so the direction between them is again an assumption.

2.2 The learned network

The second network was constructed using bnlearn's implementation of hill-climbing structure search, scored by the Bayesian Information Criterion, starting from an empty graph with no connections forced or forbidden in advance. A tabu list of length 100, a convergence tolerance of 1×10^{-4}, and a maximum of 10,000

iterations. The Bayesian Information Criterion penalises additional parameters, which was appropriate given that only 242 training records were available to estimate the network's conditional probability tables; a criterion that rewarded fit alone would risk selecting a structure that merely captured noise in the training sample. Hill-climbing was the standard greedy search strategy used for structure learning at this scale and was computationally tractable for a network of fourteen nodes.

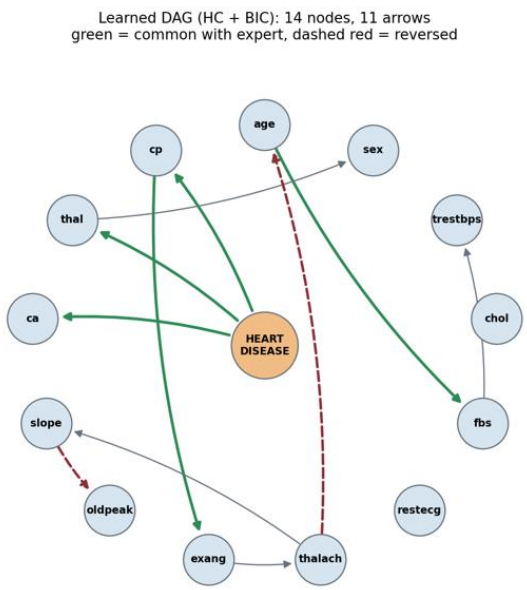


Figure 2. The learned directed acyclic graph, comprising 14 nodes and 11 directed edges. Edges shown in green also appear in the expert network; edges shown as a dashed line appear in both networks but with different directions reversed.

Starting from an empty graph, with neither edges required nor edges excluded in advance, ensured that the learned structure is not seeded by the assumptions already encoded in the expert network, so that the comparison between the two structures was an independent one rather than one biased toward agreement. The tabu list allowed the search to escape short cycles in which the same edge is repeatedly added, removed, and reintroduced. At no point in this process was the test set consulted, so the learned structure's later performance on that set constituted a genuine out-of-sample evaluation. The principal limitation of this approach was that hill-climbing is a greedy, local search: it can terminate at a local rather than a global optimum, and different runs, or different implementations and can produce structures of equal score in which shared edges are oriented differently.

2.3 Comparing the two networks

The two structures share five connections in the same direction, disagree on the direction of two further connections that appear in both, and are otherwise composed of edges unique to one network or the other, as summarised in Table 4.

Table 4. Comparison of directed sedges between the expert-designed and learned networks

Category	Count	Edges involved
Shared, same direction	5	age to fasting blood sugar; chest-pain type to exercise angina; disease to vessels affected; disease to chest-pain type; disease to thalassemia result
Shared, reversed direction	2	age to maximum heart rate in the expert network, reversed to maximum heart rate to age in the learned network; ST depression to ST slope in the expert network, reversed to ST slope to ST depression in the learned network
Present only in the expert network	14	age, blood pressure, cholesterol, fasting blood sugar and sex each to disease; disease to maximum heart rate, exercise angina, ST depression, ST slope and resting ECG; age to cholesterol; age to blood pressure; blood pressure to resting ECG; exercise angina to ST depression
Present only in the learned network	4	exercise angina to maximum heart rate; fasting blood sugar to blood pressure; thalassemia result to sex; maximum heart rate to ST slope

The learned network makes disease a root node with three children, chest-pain type, thalassemia result and vessels affected, so its Markov blanket is just those three. The Bayesian Information Criterion judged the risk-factor edges not worth their parameter cost at 242 patients, consistent with their low mutual information. The expert network retains all thirteen remaining variables in the target's blanket.

Two learned edges are hard to justify clinically: heart rate into age, and thalassemia result into sex, since a later test result cannot alter a fixed attribute. Both are better read as artefacts of an equally scoring alternative orientation than as evidence of a mechanism

The ST-depression to ST-slope reversal is unremarkable, as both summarise one exercise response. The remaining learned-only edges, fasting glucose to blood pressure and exercise angina to heart rate, more plausibly reflect shared physiology than direct causation. Overall, neither the presence nor the absence of an edge in either network proves or rules out a genuine biological mechanism.

3. Parameter Learning

The conditional probability tables for all fourteen nodes in both networks were estimated using the training data only, applying a Bayesian estimator with a BDeu prior and an equivalent sample size of five. Every node, including any node that would otherwise be isolated, was retained and given a valid table; each resulting table was checked to confirm that its entries were finite and positive and that every column summed to one. All twenty-eight tables, fourteen for each network, were exported to CSV, and both fitted networks were saved as pickle files, expert_bayesian_network.pkl and learned_bayesian_network.pkl, together with a preprocessing.pkl file recording the imputation values, discretisation cut points and variable states, so that either network can be reloaded and applied to new data without repeating the fitting procedure.

Table 5. Structural complexity of the expert and learned networks: edges, free parameters and target-node CPT support

Network	Edges	Free parameters	Target-node configurations	Configurations with no support	Configurations with fewer than 5 patients
Expert	21	194	108	39	93
Learned	11	57	1	0	0

Because the learned network gives heartdisease_bin no parents, its conditional probability table reduces to a single row: the prior probability of disease is estimated at 46.0%, based on all 242 training patients. In the expert network, the same table spans 108 rows corresponding to every combination of its five parent variables; of these, 93 rows are supported by fewer than five training patients, and 39 rows have no supporting patients at all. Those 39 empty rows default to a probability of 50% purely because of the symmetric BDeu prior, and this figure must not be interpreted as an observed disease rate.

3.2 Conditional probability tables for the target's direct parents

Chest-pain type, thalassemia result and the number of affected vessels each have heartdisease_bin as their only parent in both networks, so their conditional probability tables are directly comparable, and identical, across the two structures.

Table 6. Conditional probability tables for chest-pain type(cp), thalassemia result(thal) and vessels affected(ca), each conditioned-on disease status.

Disease status	n	P(cp=1)	P(cp=2)	P(cp=3)	P(cp=4)	P(thal=3)	P(thal=6)	P(thal=7)	P(ca=0)	P(ca=1)	P(ca=2)	P(ca=3)
No disease	131	.102	.267	.372	.259	.800	.036	.164	.806	.132	.050	.012
Disease	111	.058	.067	.138	.737	.272	.069	.659	.367	.323	.208	.102

Six patterns in the table are interpreted individually, since together they explain much of what distinguishes the two disease groups in this sample.

- Asymptomatic chest pain (cp = 4) is the clearest single marker of disease: 73.7% of diseased versus 25.9% of non-diseased patients, a difference of 47.7 percentage points.
- A reversible perfusion defect (thal = 7) shows the same pattern: 65.9% versus 16.4% (+49.6 pp); a normal result is far more common among non-diseased patients (80.0% versus 27.2%).
- Vessel involvement rises sharply with disease: P(ca=0) is 80.6% without disease versus 36.7% with; two or three vessels appear in roughly 31% of diseased but only about 6% of non-diseased patients.
- A common finding need not be diagnostic: ca = 0 remains the single most frequent state even among diseased patients (36.7%), showing that P(finding given disease) is not P(disease given finding).

- Exercise angina differs by model: P(angina given disease) is 54.0% for the expert network (a direct edge) versus 44.0% for the learned network (via chest-pain type); for non-diseased patients, 15.2% versus 23.6%.
- The expert target table thins quickly: high age, blood pressure and cholesterol in a male patient gives 99.5% (n = 5); low values in a female patient give 0.3% (n = 9); medium age with high blood pressure and cholesterol in a male gives 88.7% (n = 9). These rest on very few patients and are suggestive only.

4. Inference

Highest-probability combination: The expert network's highest estimate, 99.96%, applies to a male patient with asymptomatic chest pain, exercise angina, a reversible defect, high ST depression, low heart rate, a flat slope and three vessels affected, supported by one training patient. The learned network reaches 98.79% for any patient with asymptomatic chest pain, a reversible defect and three vessels affected, regardless of the other five variables, since none fall within its Markov blanket; no training patient matches this exact combination.

Lowest-probability combination: The expert network's lowest estimate, 0.05%, applies to a female patient with atypical angina, exercise angina, a normal thalassemia result, no ST depression, high heart rate, a downsloping segment and no vessels affected, unsupported in training. The learned network's lowest, 3.20%, needs only atypical angina, a normal result and no vessels affected, supported by one patient.

Effect of changing one variable at a time: Using the most common training profile as a baseline (supported by seven patients), the baseline probability of disease is 1.1% in the expert network and 3.2% in the learned network.Table 7 reports the effect of changing one variable at a time. In both networks, vessels affected produces the largest shift: raising it from zero to three moves the expert estimate from 1.1% to 16.8% (+15.7 pp) and the learned estimate from 3.2% to 37.9% (+34.7 pp). The learned network is unaffected by angina, heart rate or sex, since none lie in its Markov blanket; the expert network responds moderately to each.

Posterior probabilities given disease = 1: The conditional probabilities of chest-pain type, thalassemia result and vessels affected given disease are reported in Table 6 and are identical in both networks. For exercise angina given disease, the expert

network estimates 54.0% against 44.0% for the learned network, which reaches this variable indirectly through chest-pain type.

Table 7. Change in the posterior probability of disease when a single variable is altered from the baseline profile, expressed in percentage points (pp)

Variable changed from its baseline value	Expert network	Learned network
Vessels affected: 0 to 3	1.1% to 16.8% (+15.7 pp)	3.2% to 37.9% (+34.7 pp)
Vessels affected: 0 to 2	+8.1 pp	+20.2 pp
Thalassemia result: normal to reversible defect	+10.4 pp	+25.0 pp
Chest-pain type: atypical to asymptomatic	+4.2 pp	+24.0 pp
Maximum heart rate: high to low	+7.2 pp	no change (outside Markov blanket)
Exercise angina: absent to present	−0.8 pp	no change (outside Markov blanket)
Sex: male to female	−0.8 pp	no change (outside Markov blanket)

Table 8. The five highest-ranked evidence combinations in the expert network, drawn from the full set of 5,184 combinations evaluated.

Rank	Evidence combination (sex, chest pain, angina, thal, ST depression, heart rate, slope, vessels)	P(disease)	Training patients supporting it
1	Male, asymptomatic, present, reversible defect, high, low, flat, 3	99.96%	1
2	Male, asymptomatic, present, reversible defect, high, low, flat, 2	99.93%	7
3	Male, asymptomatic, present, reversible defect, high, low, flat, 1	99.88%	4
4	Male, asymptomatic, present, reversible defect, moderate, low, flat, 1	99.78%	4
5	Male, asymptomatic, present, reversible defect, high, low, upsloping, 3	99.77%	1

4.1 Comparing the inference results across the two networks

The two networks agree in direction and rank patients identically on the three shared parents, since those CPTs match exactly. They diverge in sensitivity to the remaining five variables: the expert network responds gradually to heart rate, ST depression, ST slope, sex and angina, while the learned network ignores all five, as none lie within its Markov blanket. Consequently, the learned network's estimates take fewer distinct values and move in larger steps, while the expert network's vary more smoothly.

5. Model Evaluation and reflection

Each test patient's thirteen predictors were entered as evidence and P(disease) computed at a fixed 0.50 threshold. A majority-class baseline, always predicting no disease at the training prevalence, is shown for reference in table 9.

Table 9. Each test patient's thirteen predictors were entered as evidence and P(disease) computed at a fixed 0.50 threshold.

Model	Accuracy	Precision	Recall	Specificity	F1	ROC-AUC	Average precision	Brier score	TN	FP	FN	TP
Expert network	.853	.807	.893	.818	.848	.939	.934	.105	27	6	3	25
Learned network	.869	.857	.857	.879	.857	.918	.914	.108	29	4	4	24
Majority-class baseline	.541	0	0	1	0	.500	.459	.248	33	0	28	0

When determining which network performs better it depended on which aspect of performance is prioritised. The expert network ranks higher (AUC 0.939 vs 0.918, lower Brier score) and misses one fewer case (three vs four false negatives); the learned network is more accurate, precise and specific, with two fewer false positives. A paired bootstrap of the AUC difference gives a 95% interval of −0.098 to 0.039, which includes zero, so neither is proven better; by the pre-specified primary metric, ROC-AUC, the expert model leads narrowly.

The best-performing model is not necessarily the most interpretable one. The learned network is smaller (57 vs 194 parameters), fully supported by training data, and has an easily described three-variable Markov blanket, but contains two clinically implausible edges: thalassemia result into sex, and heart rate into age.

The expert network is clinically coherent throughout, but its 108-row target table is mostly unsupported and leans heavily on the prior. The learned structure is the more statistically defensible summary of this data; the expert structure remains the more clinically readable one.

6. Conclusion

This report compared an expert-designed Bayesian network with a structure learned directly from the Cleveland heart-disease data, and both performed well

above the majority-class baseline without either proving reliably superior as a classifier: the expert network ranked patients marginally higher on ROC-AUC, while the learned network was more accurate and precise at the standard threshold, a difference not statistically distinguishable on this test set. Chest-pain type, thalassemia result and the number of vessels affected emerged as the dominant predictors in both networks, with vessel count producing the largest change in predicted risk, though the two structures diverged in interpretability: the expert network offered a fully traceable clinical rationale across all thirteen predictors, while the learned network reduced the problem to a compact three-variable core at the cost of two clinically indefensible edges, so the best-performing model was not unambiguously the most interpretable one. Taken together, the two networks corroborated one another on the clinical substance of the problem, illustrating structure learning's ability to recover a defensible, parsimonious model without contradicting established clinical reasoning and supporting their use as complementary tools; any application beyond would require validation on an independent, larger and more representative patient sample.